

F. Y. B. SC. SEMESTER II

BIOCHEMISTRY: BIOMOLECULES

S. No.	Topic	Hours (45)
1	Biomolecules: • Structure, function, diversity and distribution • General introduction of composition of living matter, Cell wall structure with reference to gram positive and gram negative bacteria Carbohydrates • Monosaccharides and their inter relationship, structure of sugar, • Stereoisomerism and optical isomerism of sugars. Ring structure and tautomeric forms, mutarotation. • Important derivatives of Monosaccharides, Disaccharides and Trisaccharides (Glucose, fructose, maltose, lactose, cellobiose, gentiobiose, Melibiose, Turanose, Sucrose, Trehalose, Mannotriose, Rabinose, Rhamnose, Raffinose, Gentionose, Melizitose.) • Structure, occurrence and biological importance of structural polysaccharides e.g. Cellulose, chitin, agar, algenic acids, pectins, glycoproteins, proteoglycans, sialic acids, blood group polysaccharides, bacterial cell wall polysaccharides.	15

S. No.	Topic	Hours (45)
2	Lipids • Building block of lipids – fatty acids, glycerol, sphingosine Definition and classification of lipids. • Classification of fatty acids, physio-chemical properties of fatty acids, separation of fatty acids, distribution of fatty acids in nature and characterization of fatty acids, saponification and iodine number. Properties of glycerol, fats and oils. • Systematic nomenclature and classes of glycerides, MAG, DAG, TG, phospholipids, PA, PG, PE, PS, LPC, PI and plasmalogens, sphingolipids - sphingosine, ceramide, sphingomyelin, glycolipids cerebrosides, gangliosides and sialic acids. • Properties and function of phospholipids and Prostaglandins. • Isoprenoids – types and structures, structure of sterols, Bile acids, steroid hormones, plant sterol, ergosterol, stigma sterol, cholesterol, glucocorticoid, mineralocorticoids. • Lipoproteins – classification, composition and their importance, Role of Lipids in cellular architecture and functions. Vitamins • Structure and function of water and fat soluble vitamins • Dietary requirements • Deficiency disorders	15

S. No.	Topic	Hours (45)
3	Proteins • Amino acids: Classification, Properties, reactions, Rare amino acids, and separation techniques • Proteins: Classification, Reactions, functions, properties, peptide synthesis, solid phase synthesis • Structure: Peptide bond, primary structure; Secondary structure: alpha helix, beta structure, β- helix, super secondary Structure; Tertiary structure Nucleic acids • Importance of nucleic acids in living system, general composition of nucleic acids, the purine and pyrimidine bases, Tautomeric forms of bases. • Reactions of purines and pyrimidines, structure of nucleosides and nucleotide, deoxynucleotides, cyclic nucleotides and polynucleotides. • Watson and crick model for DNA. Different types of DNA and RNA	15

Books

- Nelson et al., Lehninger Principles of Biochemistry, Sixth edition, 2008, W.H Freeman and Company.
- Voet and Voet, Biochemistry, fourth edition, 2011, John Wiley and sons.
- Berg et al., Biochemistry, fifth edition, 2009, W.H Freeman and Company.
- Horton et al., Principles of Biochemistry, 5th Edition, 2011, Pearson publishing group.

THE ORIGIN OF LIFE

- Radioactive dating studies indicate that Earth was formed ~4.6 billion years ago but, due to the impacts of numerous large objects, its surface remained too hot to support life for several hundred million years thereafter.
- The earliest evidence of cellular life, microfossils of what appear to be organisms resembling modern cyanobacteria, is ~3.5 billion years old.
- It is generally accepted that the development of life occupied three stages:
 1. Chemical evolution, in which simple geologically occurring molecules reacted to form complex organic polymers. [Oparin & Haldane, 1920s; Miller and Urey, 1953]
 2. The self-organization of collections of these polymers to form replicating entities. At some point in this process, the transition from a lifeless collection of reacting molecules to a living system occurred.
 3. Biological evolution to ultimately form the complex web of modern life.

Distinguishing features of living organisms

- A high degree of chemical complexity and microscopic organization.
- Systems for extracting, transforming, and using energy from the environment
- Defined functions for each of an organism's components and regulated interactions among them.
- Mechanisms for sensing and responding to alterations in their surroundings
- A capacity for precise self-replication and self-assembly
- A capacity to change over time by gradual evolution

THE CELL

- The cell is the structural and functional unit of life.
- It may be also regarded as the **basic unit of biological activity**.
- *A living cell is a true representative of life with its own organization and specialized functions.*
- The cells of the living kingdom may be divided into two categories **Prokaryotic and eukaryotic cells**

1. Prokaryotes

- (Greek : pro – before; karyon –nucleus) lack a well defined nucleus and possess relatively simple structure. These include the various bacteria.

2. Eukaryotes

- (Greek : eu – true; karyon – nucleus) possess a well defined nucleus and are more complex in their structure and function. The higher organisms (animals and plants) are composed of eukaryotic cells.

Comparison between prokaryotic and eukaryotic cells

<i>Characteristic</i>	<i>Prokaryotic cell</i>	<i>Eukaryotic cell</i>
1. Size	Small (generally 1-10 μm)	Large (generally 10-100 μm)
2. Cell membrane	Cell is enveloped by a rigid cell wall	Cell is enveloped by a flexible plasma membrane
3. Sub-cellular organelles	Absent	Distinct organelles are found (e.g. mitochondria, nucleus, lysosomes)
4. Nucleus	Not well defined; DNA is found as nucleoid, histones are absent	Nucleus is well defined, surrounded by a membrane; DNA is associated with histones
5. Energy metabolism	Mitochondria absent, enzymes of energy metabolism bound to membrane	Enzymes of energy metabolism are located in mitochondria
6. Cell division	Usually fission and no mitosis	Mitosis
7. Cytoplasm	Organelles and cytoskeleton absent	Contains organelles and cytoskeleton (a network of tubules and filaments)

EUKARYOTIC CELL

- The human body is composed of about 10^{14} cells.
- There are about 250 types of specialized cells in the human body e.g. erythrocytes, nerve cells, muscle cells, β cells of pancreas.
- The plant cell differs from an animal cell by possessing a rigid cell wall (mostly composed of cellulose) and chloroplasts. The latter are the sites of photosynthesis.
- The cell consists of well defined subcellular organelles, enveloped by a plasma membrane.

Nucleus

- Nucleus is the largest cellular organelle, surrounded by a double membrane nuclear envelope.
- The outer membrane is continuous with the membranes of endoplasmic reticulum.
- At certain intervals, the two nuclear membranes have nuclear pores with a diameter of about 90 nm. These pores permit the free passage of the products synthesized in the nucleus into the surrounding cytoplasm.
- Nucleus contains **DNA**, the repository of genetic information.
- The nucleus of the eukaryotic cell contains a dense body known as **nucleolus**. It is rich in RNA, particularly the ribosomal RNA which enters the cytosol through nuclear pores.
- The ground material of the nucleus is often referred to as **nucleoplasm**. It is rich in enzymes such as DNA polymerases and RNA polymerases

Mitochondria

- The mitochondria (Greek: mitos – thread; chondros – granule) are the centres for the cellular respiration and energy metabolism.
- They are regarded as the **power house of the cell** with variable size and shape, as they are **principal producers of ATP** in the aerobic cells.
- Mitochondria are rod-like or filamentous bodies, usually with dimensions of $1.0 \times 3 \mu\text{m}$. About 2,000 mitochondria, occupying about 1/5th of the total cell volume, are present in a typical cell.
- The internal chamber of mitochondria is referred to as **matrix** or **mitosol**.
- The components of electron transport chain and oxidative phosphorylation are buried in the inner mitochondrial membrane. The matrix contains several enzymes concerned with the energy metabolism
- The mitochondrial matrix contains a circular double stranded DNA (**mtDNA**), RNA and ribosomes.

Endoplasmic reticulum

- The network of membrane enclosed spaces that extends throughout the cytoplasm constitutes endoplasmic reticulum (ER).
- A large portion of the ER is studded with ribosomes to give a granular appearance which is referred to as **rough endoplasmic reticulum**.
- During the process of cell fractionation, rough ER is disrupted to form small vesicles known as **microsomes**.
- The smooth endoplasmic reticulum does not contain ribosomes. It is involved in the synthesis of lipids (triacylglycerols, phospholipids, sterols) and metabolism of drugs, besides supplying Ca^{2+} for the cellular functions.

Golgi apparatus

- Eukaryotic cells contain a unique cluster of **membrane vesicles** known as **dictyosomes** which, in turn, constitute Golgi apparatus (or Golgi complex).
- The newly synthesized proteins are handed over to the Golgi apparatus which catalyse the addition of carbohydrates, lipids or sulfate moieties to the proteins.
- Golgi apparatus are also involved in the **membrane synthesis**, particularly for the formation of intracellular organelles (e.g. peroxisomes, lysosomes).

GOLGI BODY

- The Golgi apparatus is an organelle in eukaryotic organisms that moves molecules from the endoplasmic reticulum to their destination. The organelle also modifies products of the endoplasmic reticulum to their final form. The Golgi apparatus is comprised of a series of flattened sacs that extend from the endoplasmic reticulum.

MAJOR FUNCTIONS OF GOLGI BODY:

- The Golgi has important functions in tagging vesicles with proteins and sugar molecules, which serve as identifiers for the vesicles so they can be delivered to the proper target.
- All functions are associated with moving molecules from the endoplasmic reticulum to their final destination and modifying certain products along the way.
- The multiple sacs of the Golgi serve as different chambers for chemical reactions. As the products of the endoplasmic reticulum move through the Golgi apparatus, they are continuously transferred into new environments, and the reactions that can take place are different. So this way multiple products can be combined to form large macromolecules.

FUNCTIONS OF MULTIPLE SACS IN GOLGI BODY :

- The many sacs and folds of the Golgi apparatus allow for many reactions to take place at the same time, increasing the speed at which an organism can produce products.
- The many sacs of the Golgi apparatus function to provide many different areas in which reactions can take place in the most favorable of conditions.

GOLGI BODY

ROLE OF GOLGI BODY IN STOMACH:

- The acid is produced by reactions in the endoplasmic reticulum and is modified as it goes through the Golgi apparatus. Once to the trans side of the Golgi apparatus, the acid is packaged in a vesicle and sent towards the cell's surface. As the vesicle joins with the plasma membrane, the acid is released into the stomach.

THEORY OF GOLGI APPARATUS FUNCTION

- The most prevalent theory of how the Golgi apparatus forms is the cisternal maturation model. This model suggests that the sacs themselves tend to move from the cis face to the trans face of the Golgi apparatus over time. New sacs are formed closest to the endoplasmic reticulum. These sacs “age” as they move towards the trans face of the Golgi apparatus and their product becomes fully mature.

KEY FUNCTIONS OF GOLGI BODY:

- The Golgi apparatus and endoplasmic reticulum work together to produce new cell membrane, as well as recycle the cell membranes of vesicles by merging two membranes when vesicles are absorbed.
- The Golgi also creates lysosomes
- GOLGI BODY FUNCTIONS IN THE PLANTS:
- In plants, Golgi body is used to synthesise polysaccharide and it is also used to form large vacuole. However, in plants Golgi body does NOT form lysosomes.

Lysosomes

- Lysosomes are spherical vesicles enveloped by a single membrane.
- They are actively involved in digestion of cellular substances—namely proteins, lipids, carbohydrates and nucleic acids.
- The lysosomal enzymes are responsible for **maintaining the cellular compounds in a dynamic state**, by their degradation and recycling.

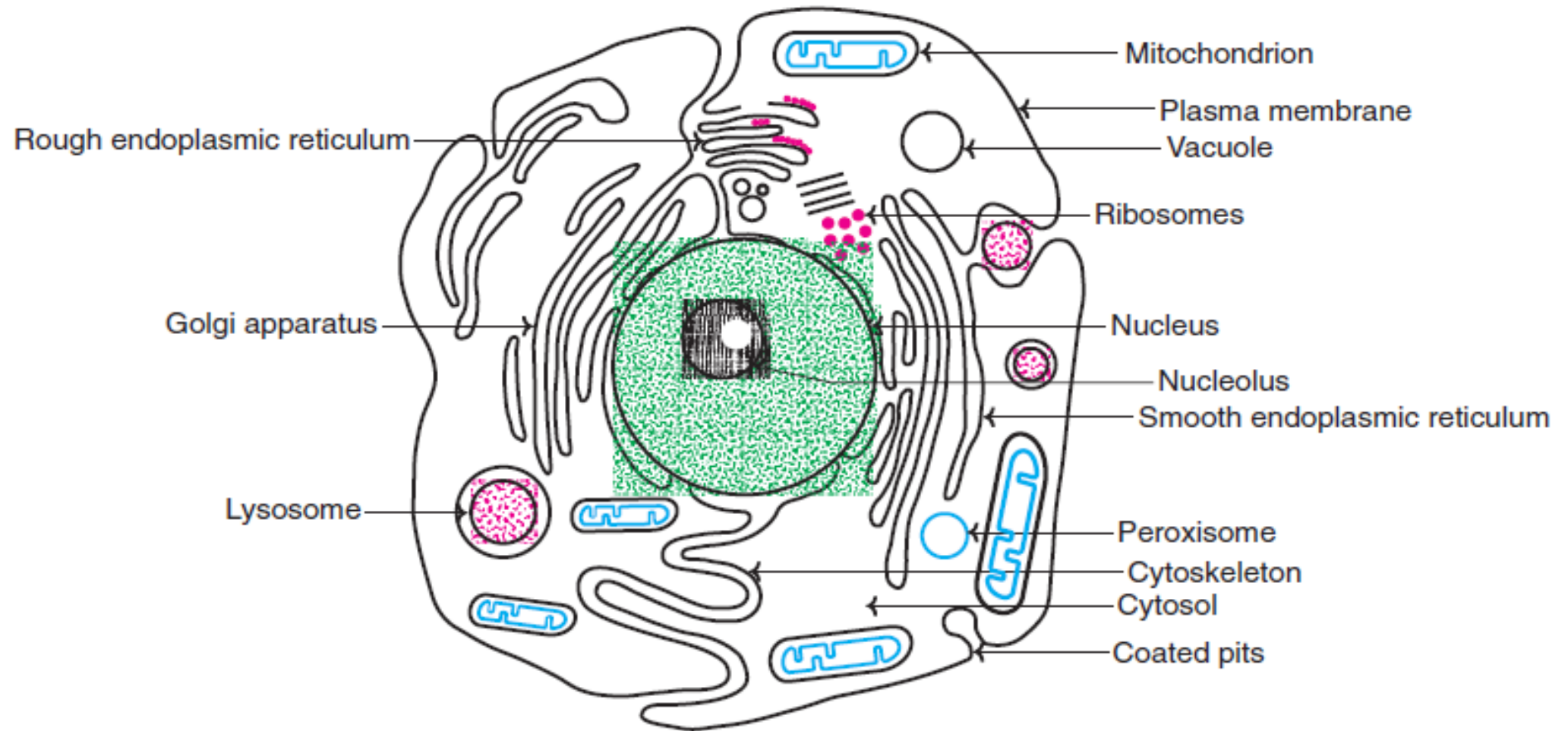
Peroxisomes

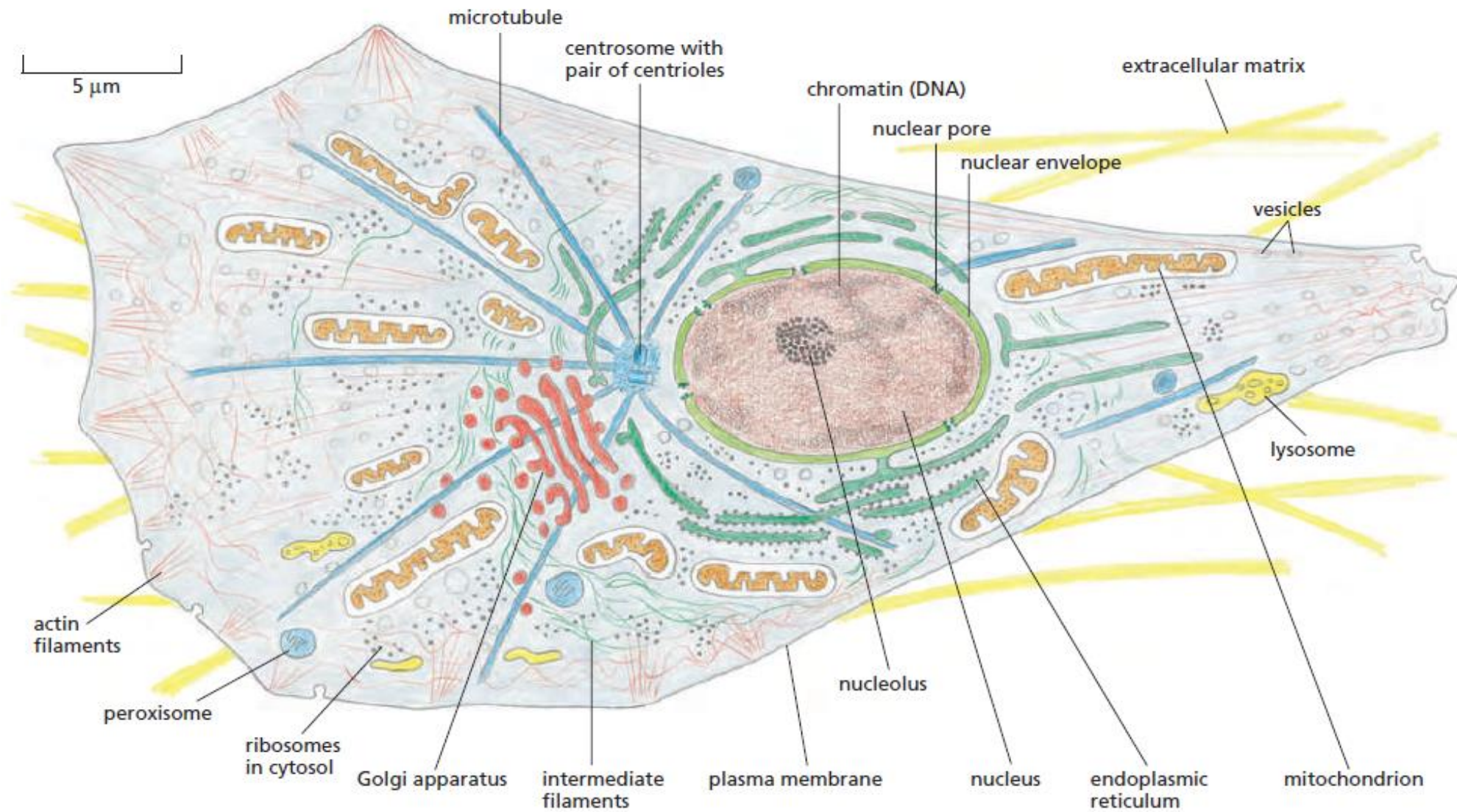
- Peroxisomes, also known as **microbodies**, are single membrane cellular organelles.
- They are spherical or oval in shape and contain the enzyme **catalase**. Catalase protects the cell from the toxic effects of H_2O_2 by converting it to H_2O and O_2 .
- Peroxisomes are also involved in the oxidation of long chain fatty acids (> C18), and synthesis of certain phospho- and glyco-lipids.
- Plants contain **glyoxysomes**, a specialized type of peroxisomes, which are involved in the glyoxylate pathway

Cytosol and cytoskeleton

- The **cellular matrix** is collectively referred to as cytosol. Cytosol is basically a compartment containing several enzymes, metabolites and salts in an aqueous gel like medium.
- The cytoplasm contains a complex network of protein filaments, spread throughout, that constitutes cytoskeleton.
- The cytoplasmic filaments are of three types – **microtubules**, actin filaments and intermediate filaments.
- The filaments which are polymers of proteins are responsible for the structure, shape and organization of the cell

Diagrammatic representation of a rat liver cell.





a typical animal cell

Biomolecules: Structure function, diversity and distribution

Chemical molecules of life

- Life is composed of **lifeless chemical molecules**
- Biochemistry shows how the **collection of inanimate molecules** that constitute living organisms **interact to maintain and perpetuate life**
- A collection of various small organic molecules are **dissolved in the cell cytosol**
- They are the **central metabolites in major pathways** that have been **conserved** throughout the **course of evolution**

Biomolecules: Structure function, diversity and distribution

Chemical molecules of life

- The chemistry of living organisms is organized around carbon, which accounts for **more than half the dry weight of cells**
- All living organisms, from microbes to mammals, are composed of chemical substances from both the inorganic and organic world, that appear in **roughly the same proportions**, and perform the same general tasks.
- A single cell of the bacterium, **Escherichia coli** contains about **6,000** different organic compounds.
- It is believed that **man** may contain about **100,000** different types of molecules although only a few of them have been characterized.

Biomolecules: Structure function, diversity and distribution

Chemical molecules of life

- The living matter is composed of mainly six elements—**carbon, hydrogen, oxygen, nitrogen, phosphorus** and **sulfur**.
- When combined in various ways, form virtually all known organic biomolecules.
- They are initially utilized in the synthesis of a small number of building blocks that are, in turn, used in the construction of a vast array of vital macromolecules
- Several other functionally important elements are also found in the cells.
 - These include Ca, K, Cl, Na, Mg,
 - In trace amounts: B, F, Si, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Se, Mo, Sn, and I.

Elemental Composition of the Human Body

Element	Dry Weight (%) ^a	Elements Present in Trace Amounts
C	61.7	B
N	11.0	F
O	9.3	Si
H	5.7	V
Ca	5.0	Cr
P	3.3	Mn
K	1.3	Fe
S	1.0	Co
Cl	0.7	Ni
Na	0.7	Cu
Mg	0.3	Zn
		Se
		Mo
		Sn
		I

^aCalculated from Frieden, E., *Sci. Am.* **227**(1), 54–55 (1972).

Periodic table in which the 26 elements utilized by living systems are highlighted in blue

1 H	Major elements Trace elements																2 He		
3 Li	4 Be												5 B	6 C	7 N	8 O	9 F	10 Ne	
11 Na	12 Mg												13 Al	14 Si	15 P	16 S	17 Cl	18 Ar	
19 K	20 Ca			21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr			39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57–70 Lanthanides	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88 Ra	89–102 Actinides	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rt	112 Uub							

- There is no known biological requirement for 64 of the 90 naturally occurring elements
- With the exceptions of oxygen and calcium, the biologically most abundant elements are but minor constituents of Earth's crust (the most abundant components of which are O, 47%; Si, 28%; Al, 7.9%; Fe, 4.5%; and Ca, 3.5%)

Chemical Composition of Living Cells

- There are four general classes of macromolecules within living cells: **nucleic acids, proteins, polysaccharides, and lipids**.
- Although subtle differences do exist between cells (e.g., erythrocyte, liver, muscle or fat cell), they all generally contain a greater variety of proteins than any other type of macromolecule, with **about 50% of the solid matter of the cell being protein (15% on a wet weight basis)**.
- Cells generally contain many more protein molecules than DNA molecules, yet **DNA** is typically the largest biomolecule in the cell.
- About **99%** of cellular molecules are **water** molecules, with water normally accounting for approximately 70% of the total wet-weight of the cell.
- Although water is obviously important to the vitality of all living cells, the bulk of our attention is usually focused on the **other 1% of biomolecules**.

Chemical Composition of Living Cells

- These compounds, which have molecular weights ranging from 1×10^3 to 1×10^6 , are created through polymerization of building blocks that have molecular weights in the range of 50 to 150.
- Most biomolecules can be regarded as **derivatives of hydrocarbons**, with hydrogen atoms replaced by a variety of functional groups
- Typical of these are alcohols, which have one or more hydroxyl groups; amines, with amino groups; aldehydes and ketones, with carbonyl groups; and carboxylic acids, with carboxyl groups
- The collection of molecules includes the common **amino acids, nucleotides, sugars** (and their phosphorylated derivatives), and **mono, di, and tricarboxylic acids**

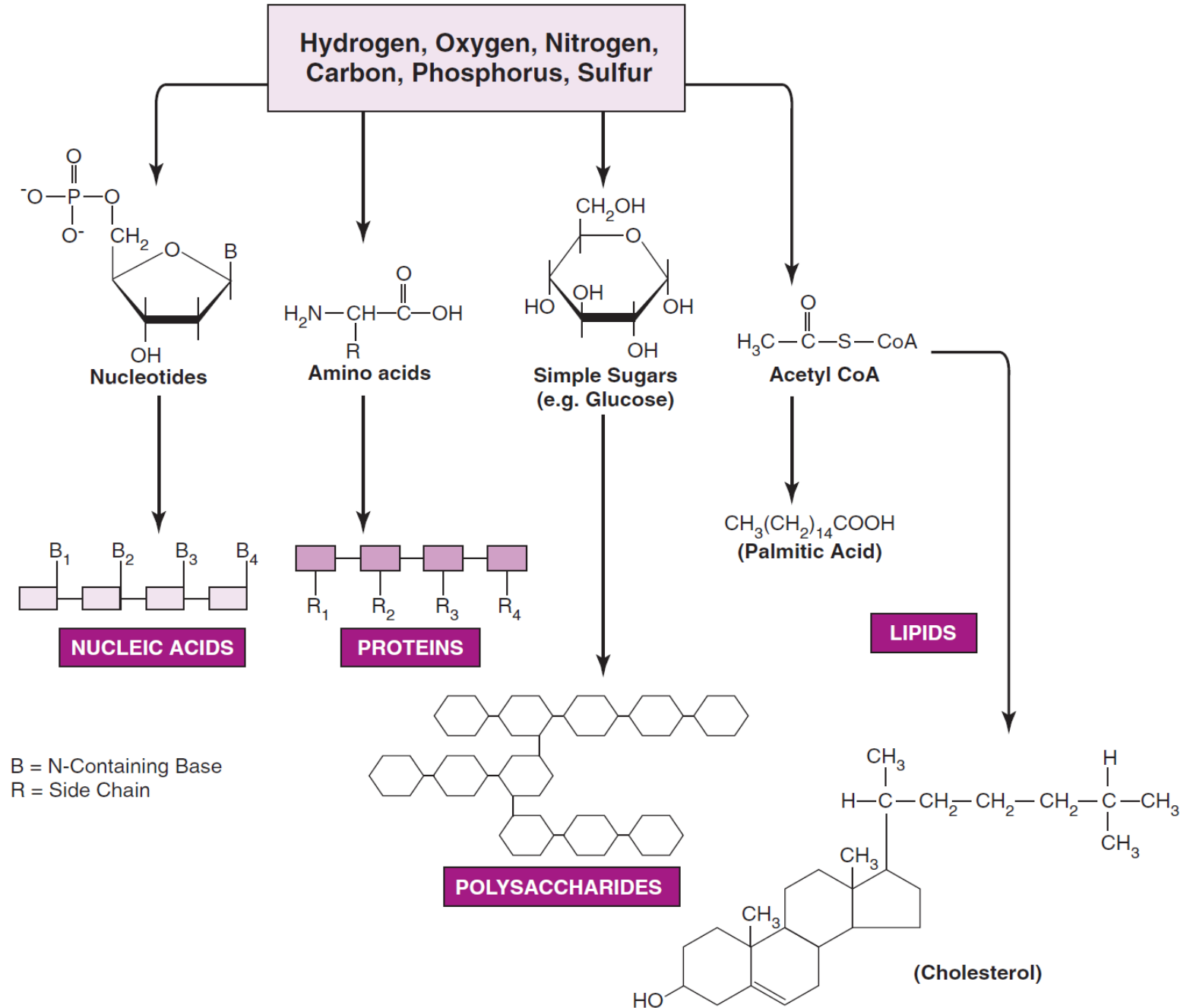
Chemical Composition of Living Cells

- These macromolecules are trapped in the cell because the **plasma membrane is impermeable** to them, although specific membrane transporters can catalyze their movement
- The entire collection of small molecules in a given cell has been called that cell's **metabolome**
- **Proteins**, long polymers of amino acids, constitute the **largest fraction** (besides water) of a cell
- The sum of all the proteins functioning in a given cell is the cell's **proteome**

Chemical Composition of Living Cells

- Some proteins have catalytic activity and function as **enzymes**; others serve as **structural elements, signal receptors, or transporters** that carry specific substances in to or out of cells
- The **nucleic acids**, DNA and RNA, are polymers of nucleotides. They store and **transmit genetic information**, and some RNA molecules have structural and catalytic roles in supramolecular complexes
- The **polysaccharides**, polymers of simple sugars such as glucose, have three major functions: as **energy-rich fuel stores**, as rigid **structural** components of cell walls (in plants and bacteria), and as **extracellular recognition** elements that bind to proteins on other cells
- The **lipids, water insoluble hydrocarbon derivatives**, serve as **structural** components of membranes, **energy-rich fuel stores, pigments, and intracellular signals**

Formation of Macromolecules Within Cells



Biomolecules Diversity in a Rapidly Dividing E. coli Cell

Material	Different Kinds of Molecules/Cell
Water	1
Nucleic acids	
DNA	1
RNA	
Ribosomal	3
Transfer	40
Messenger	1000
Nucleotides and metabolites	200
Proteins	2000-3000
Amino Acids and metabolites	100
Polysaccharides (Carbohydrates and metabolites)	200
Lipids and metabolites	50
Inorganic ions (major minerals and trace elements)	20
Others	200

Approximate Chemical Compositions of a Typical Bacterium and a Typical Mammalian Cell

COMPONENT	PERCENT OF TOTAL CELL WEIGHT	
	<i>E. COLI</i> BACTERIUM	MAMMALIAN CELL
H ₂ O	70	70
Inorganic ions (Na ⁺ , <u>K</u> ⁺ , Mg ²⁺ , Ca ²⁺ , Cl ⁻ , etc.)	1	1
Miscellaneous small metabolites	3	3
Proteins	15	18
<u>RNA</u>	6	1.1
<u>DNA</u>	1	0.25
Phospholipids	2	3
Other lipids	-	2
Polysaccharides	2	2
Total cell volume	$2 \times 10^{-12} \text{ cm}^3$	$4 \times 10^{-9} \text{ cm}^3$
Relative cell volume	1	2000

Chemical composition of a normal man (weight 65 kg)

Constituent	Percent (%)	Weight (kg)
Water	61.6	40
Protein	17.0	11
Lipid	13.8	9
Carbohydrate	1.5	1
Minerals	6.1	4

Complex biomolecules of cells

- The organic compounds such as amino acids, nucleotides and monosaccharides serve as the **monomeric units** or building blocks of complex biomolecules—proteins, nucleic acids (DNA and RNA) and polysaccharides, respectively.
- As regards to lipids, it may be noted that they are not biopolymers in a strict sense, but majority of them contain fatty acids.

Biomolecule	Building block (repeating unit)	Major functions
1. Protein	Amino acids	Fundamental basis of structure and function of cell (static and dynamic functions).
2. Deoxyribonucleic acid (DNA)	Deoxyribonucleotides	Repository of hereditary information.
3. Ribonucleic acid (RNA)	Ribonucleotides	Essentially required for protein biosynthesis.
4. Polysaccharide (glycogen)	Monosaccharides (glucose)	Storage form of energy to meet short term demands.
5. Lipid	Fatty acids, glycerol	Storage form of energy to meet long term demands; structural components of membranes.

Nucleic Acids

- Nucleic acids are nucleotide polymers (from the Greek word **poly**, meaning "several", and **mer**, meaning "unit"), that store and transmit genetic information.
- Only 4 different nucleotides are used in nucleic acid biosynthesis.
- Genetic information contained in nucleic acids is stored and replicated in **chromosomes**, which contain **genes** (from the Greek word **gennan**, meaning "to produce").
- The total number of genes in any given mammalian cell may total several thousand.
- When a cell replicates itself, identical copies of DNA molecules are produced, therefore the hereditary line of descent is conserved, and the genetic information carried on DNA is available to direct the occurrence of virtually all chemical reactions within the cell.

Nucleic Acids

- The bulk of genetic information carried on DNA provides instructions for the assembly of virtually every protein molecule within the cell.
- The flow of information from nucleic acids to protein is commonly represented as

DNA → messenger ribonucleic acid (mRNA) → transfer RNA (tRNA)
→ ribosomal RNA (rRNA) → protein,

- which indicates that the nucleotide sequence in a gene of DNA specifies the assembly of a nucleotide sequence in an mRNA molecule, which in turn directs the assembly of the amino acid sequence in protein through a tRNA and rRNA molecules.

Proteins

- Proteins are amino acid polymers responsible for implementing instructions contained within the genetic code
- **Twenty different amino acids** are used to synthesize proteins, about half are formed as metabolic intermediates, while the remainder must be provided through the diet ("**essential**" amino acids)
- Each protein formed in the body, unique in its own structure and function, participates in processes that characterize the individuality of cells, tissues, organs, and organ systems
- A typical cell contains thousands of different proteins, each with a different function, and many serve as enzymes that catalyze (or speed) reactions

Proteins

- Virtually every reaction in a living cell requires an enzyme.
- Transport different compounds either outside or inside cells {e.g., lipoproteins and transferrin (an iron-binding protein) in plasma, or bilirubin binding proteins in liver cells}
- Act as storage proteins (e.g., myoglobin binds and stores O₂ in muscle cells)
- Act as defense proteins in blood or on the surface of cells (e.g., clotting proteins and immunoglobulins)
- Act as contractile proteins (e.g., the actin, myosin and troponin of skeletal muscle fibers)
- Are merely structural in nature (e.g., collagen and elastin)
- Proteins, unlike glycogen and triglyceride, are usually not synthesized and stored as non-functional entities.

Polysaccharides

- Polysaccharides are polymers of simple sugars (i.e., monosaccharides). (The term saccharide is derived from the Greek word **sakchar**, meaning "sugar or sweetness".)
- Some polysaccharides are **homogeneous polymers** that contain only one kind of sugar (e.g., glycogen), while others are complex **heterogenous polymers** that contain 8-10 types of sugars.
- In contrast to heterogenous polymers (e.g., proteins, nucleic acids, and some polysaccharides), homogenous polymers are considered to be "**non-informational**".
- Polysaccharides, therefore, can occur as functional and structural components of cells (e.g., glycoproteins and glycolipids), or merely as non-informational storage forms of energy (e.g., glycogen).
- The 8-10 monosaccharides that become the building blocks for heterogenous polysaccharides can be synthesized from glucose, or formed from other metabolic intermediates.

Lipids

- Lipids (from the Greek word **lipos**, meaning "fat") are naturally occurring, nonpolar substances that are mostly insoluble in water, yet soluble in nonpolar solvents (like chloroform and ether).
- They serve as
 - membrane components (cholesterol, glycolipids and phospholipids),
 - storage forms of energy (triglycerides),
 - precursors to other important biomolecules (fatty acids),
 - insulation barriers (neutral fat stores),
 - protective coatings to prevent infection and excessive gain or loss of water, and some vitamins (A, D, E, and K) and hormones (steroid hormones).
- Major classes of lipids are the
 - Saturated and unsaturated fatty acids (short, medium, and long-chain),
 - Triglycerides
 - Lipoproteins {i.e., chylomicrons (CMs), very low density (VLDL), low density (LDL), intermediate density (IDL), and high density lipoproteins (HDL)},
 - Phospholipids and glycolipids,
 - Steroids (cholesterol, progesterone, etc.)
 - Eicosanoids (prostaglandins, thromboxanes, and leukotrienes).

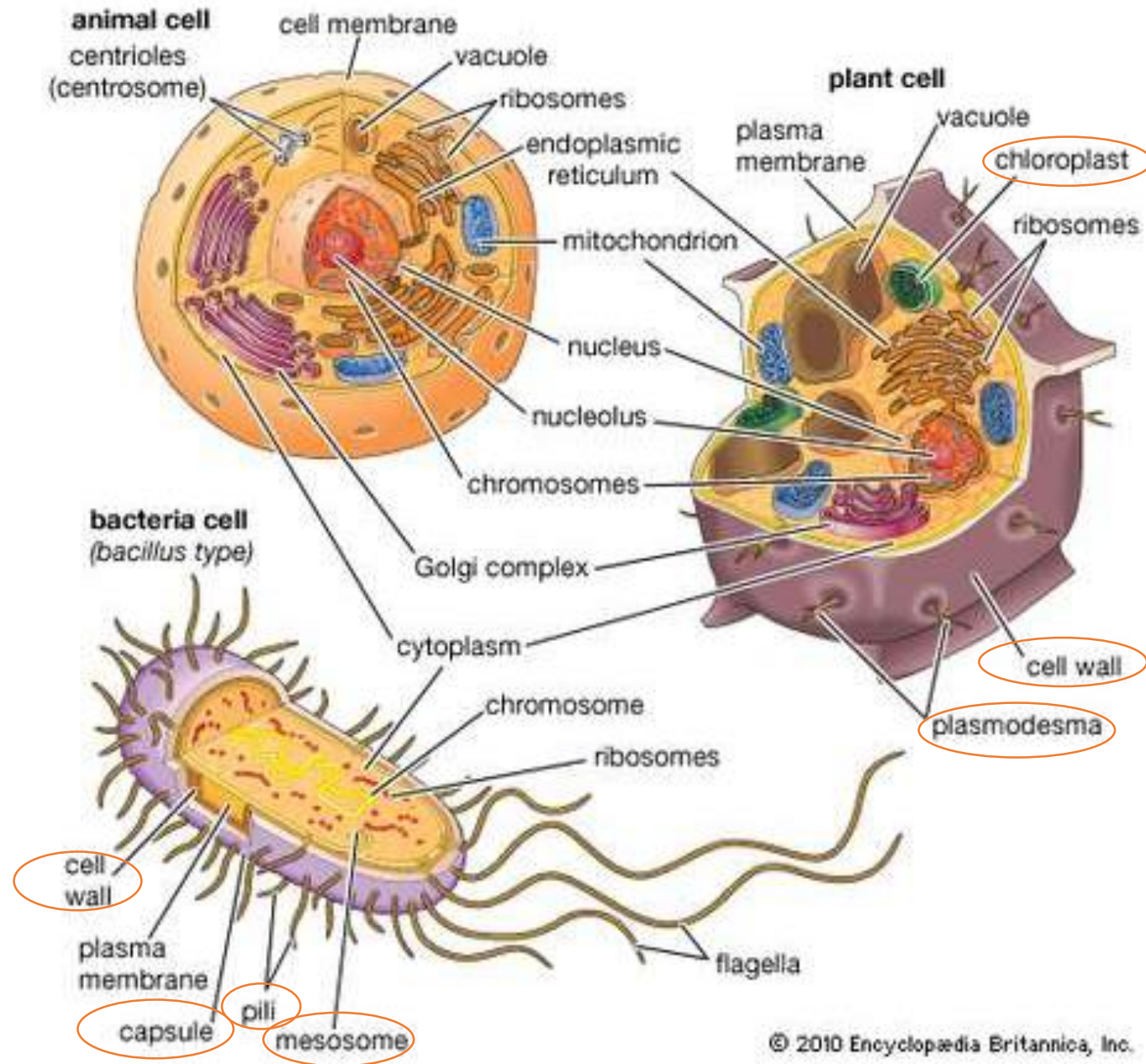
Lipids

- All lipids can be synthesized from **acetyl-CoA**, which in turn can be generated from numerous different sources, including carbohydrates, amino acids, short-chain volatile fatty acids (e.g., acetate), ketone bodies, and fatty acids.
- **Simple lipids** include only those that are esters of fatty acids and an alcohol (e.g., mono-, di- and triglycerides).
- **Compound lipids** include various materials that contain other substances in addition to an alcohol and fatty acid (e.g., phosphoacylglycerols, sphingomyelins, and cerebrosides)
- **Derived lipids** include those that cannot be neatly classified into either of the above (e.g., steroids, eicosanoids, and the fat-soluble vitamins).
- Several "**essential**" nutrients, like certain **amino acids** and **unsaturated fatty acids**, must be supplied in the diet.

Inorganic elements

- Although the study of physiological chemistry emphasizes organic molecules, the **inorganic elements** (sometimes subdivided into macrominerals, trace elements, and ultra trace elements), are also important.
- Inorganic elements are typically present in cells as ionic forms, existing as either free ions or complexed with organic molecules.
- Many "**trace elements**" are known to be essential for life, health, and reproduction, and have well-established actions (e.g., cofactors for enzymes, sites for binding of oxygen (in transport), and structural components of non-enzymatic macromolecules}

Bacterial Cell wall Structure



Types of cell wall

1. Gram positive cell wall

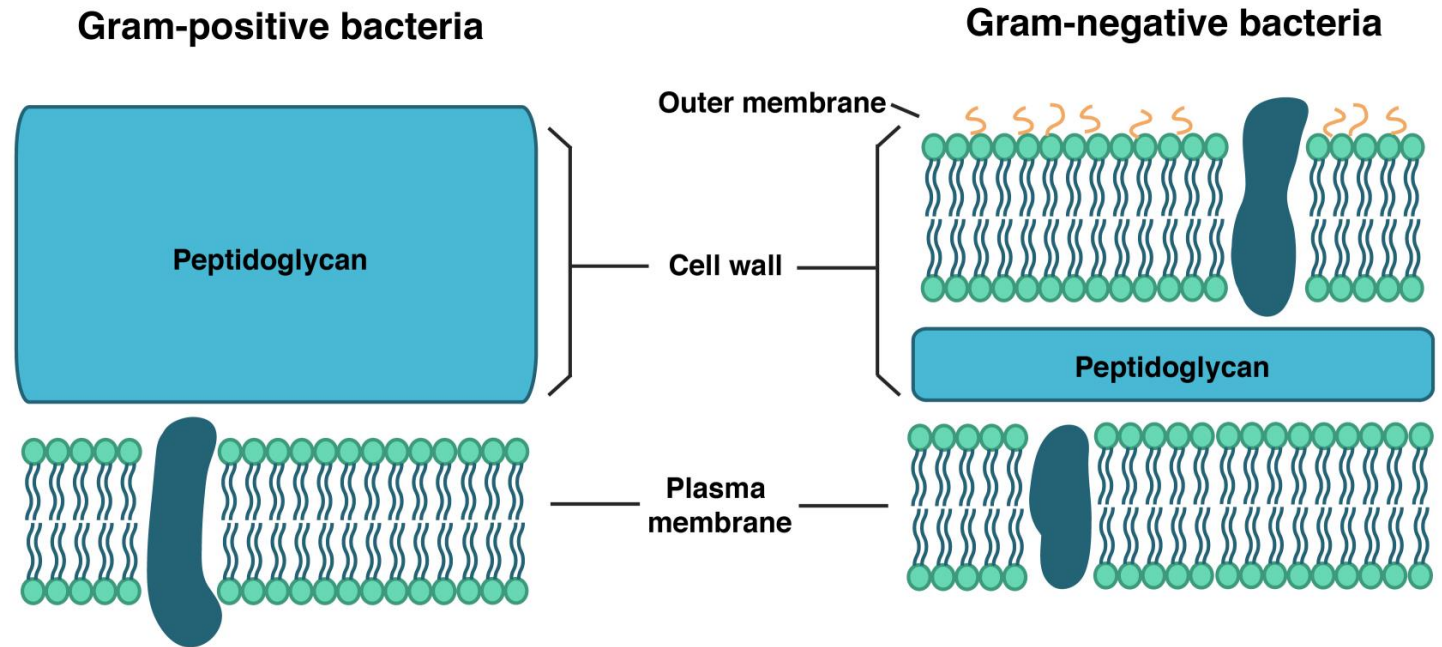
Cell wall composition of gram positive bacteria.

- Peptidoglycan
- Lipid
- Teichoic acid

2. Gram negative cell wall

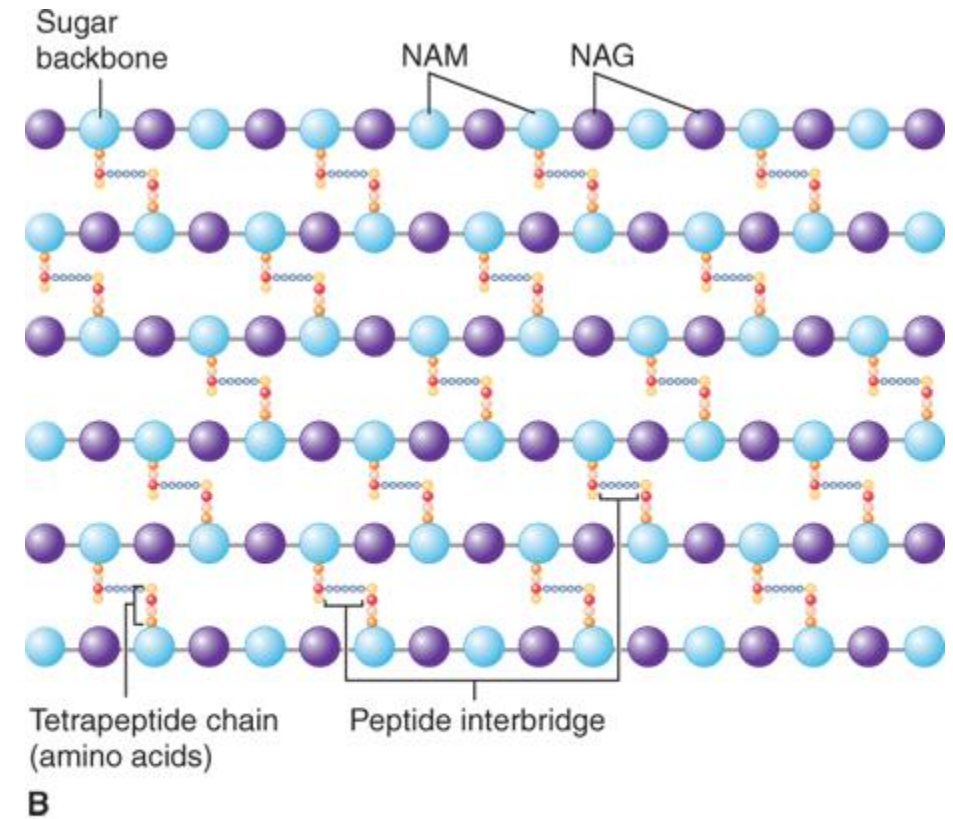
Cell wall composition of gram negative bacteria

- Peptidoglycan
- Outermembrane:
 - Lipid
 - Protein
- Lipopolysaccharide (LPS)



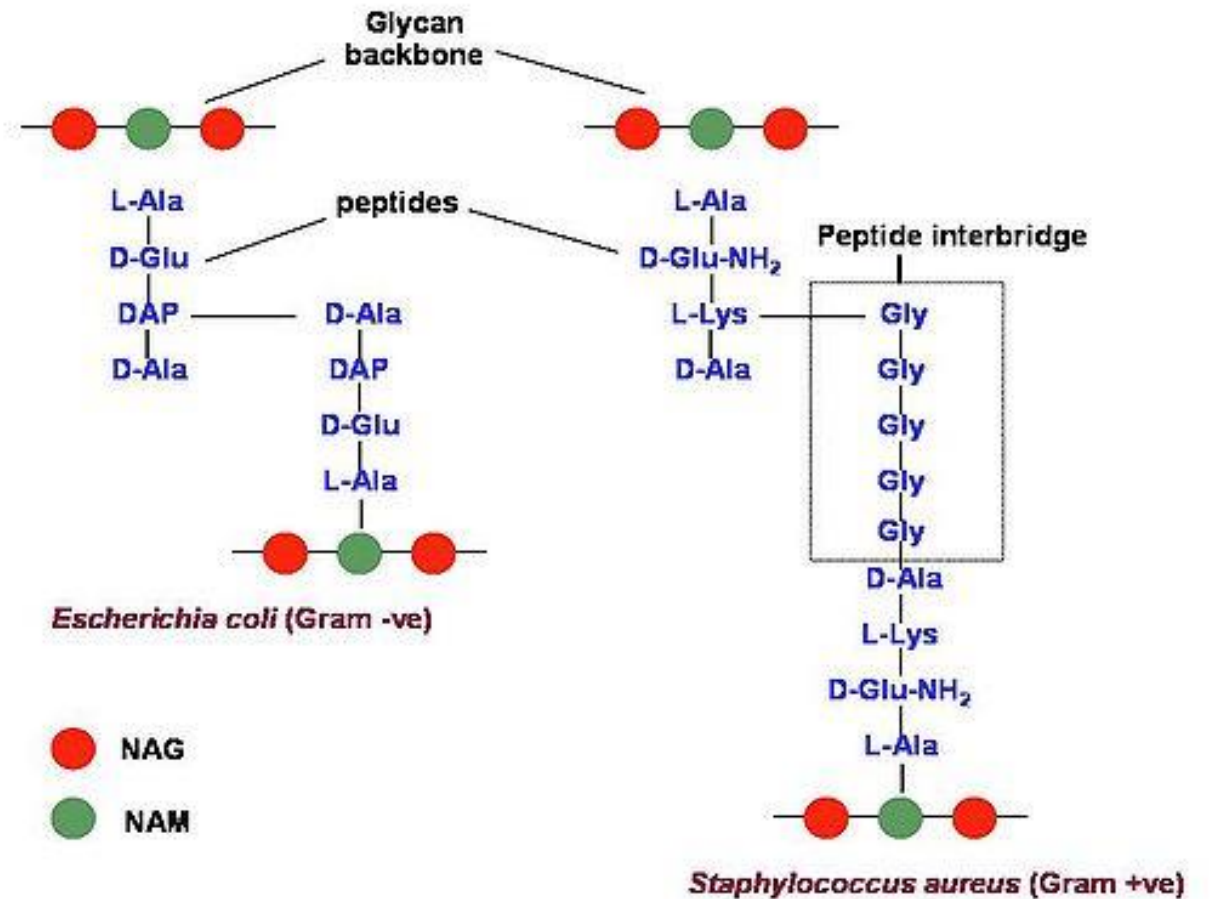
Composition of cell wall: Peptidoglycan:

- Peptidoglycan is porous cross linked polymer which is responsible for strength of cell wall. Peptidoglycan is composed of three components.
 - Glycan backbone
 - Tetra-peptide side chain (chain of 4 amino acids) linked to NAM
 - Peptide cross linkage
- Glycan backbone is the repeated unit of N-acetyl muramic acid (NAM) and N-acetyl glucosamine (NAG) linked by β -glycosidic bond.
- The glycan backbone are cross linked by tetra-peptide linkage. The tetra-peptide are only found in NAM.
- Although the peptidoglycan chemistry vary from organism to organism the glycan backbone, i.e. NAG-NAM is same in all species of bacteria.



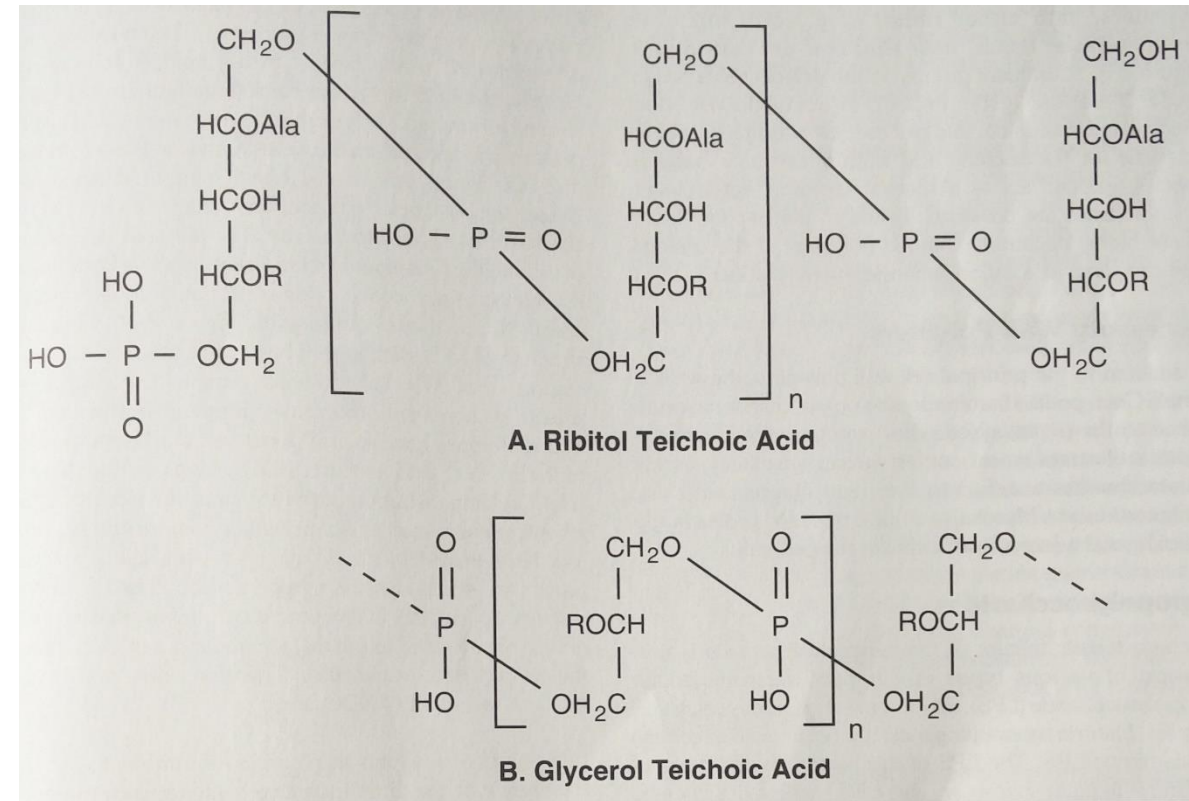
Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
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- In gram negative bacteria, peptide cross linkage occur between Diaminopimelic acid (3rd position) of one glycan back bone and D-alanine of adjacent glycan back bone.
- In gram positive bacteria, peptide cross linkage occur by peptide interbridge. The type and number of aminoacids in interbridge vary among bacterial species.



Teichoic acid:

- Teichoic acid is water soluble polymer of glycerol or ribitol phosphate.
- It is present in gram positive bacteria.
- It constitutes about 50% of dry weight of cell wall.
- It is the major surface antigen of gram positive bacteria.

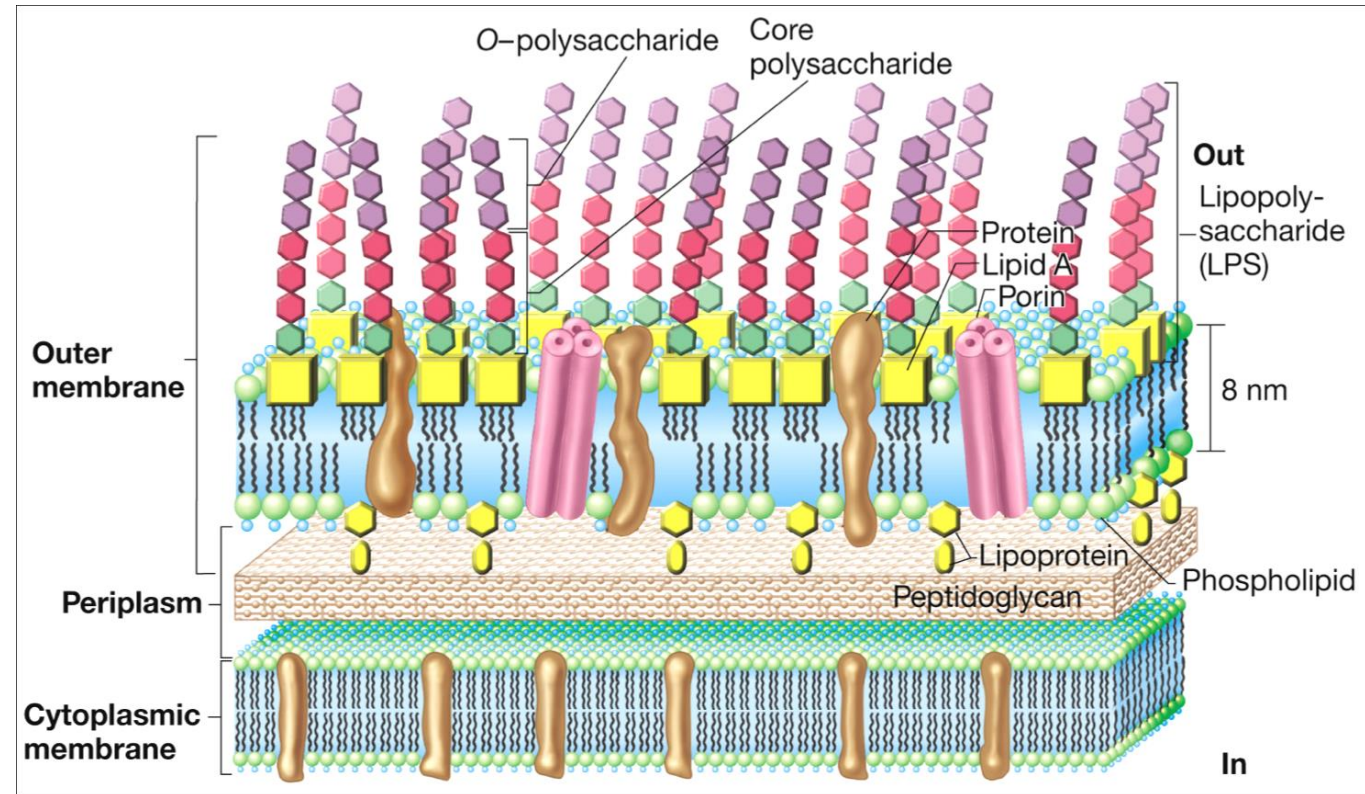


Outer membrane

- It is an additional layer present in gram negative bacteria.
- It is composed of lipid bilayer, protein and lipopolysaccharide (LPS) layer

Function of outer membrane:

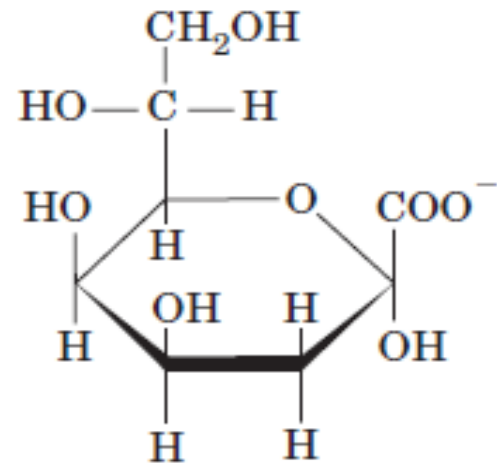
- Structure component of gram-ve cell wall
- LPS is an endotoxin produced by gram –ve bacteria
- Lipid-A is antigenic



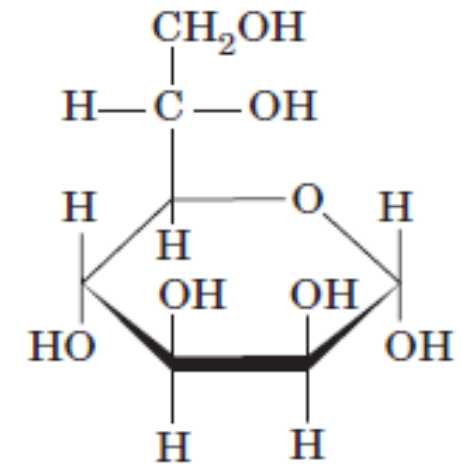
LPS

- LPS is attached to outer membrane by hydrophobic bond. LPS is synthesized in cytoplasmic membrane and transported to outer membrane.
- The LPS of the outer membrane is made up of a large complex molecules.
- They consist of lipids and carbohydrates.
- There are three major components of LPS: Lipid A, Core polysaccharide, and the O-polysaccharide.
- Lipid A is the lipid portion of the LPS which is embedded in the top layer of the outer membrane. It is phosphorylated glucosamine disaccharide.
- The core polysaccharide is attached to lipid A and consists of unusual sugars, providing stability.
- The O-polysaccharide extends outwards from the core polysaccharide and sugar molecules. It acts as an antigen and is useful for the identification of the species.

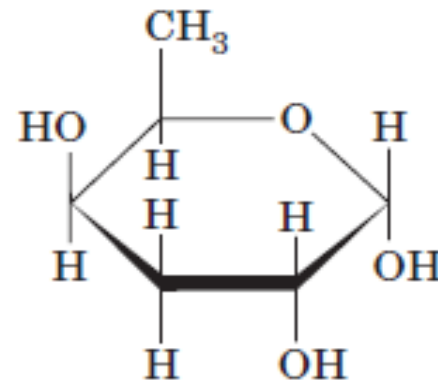
Some of the unusual
monosaccharides that occur in the
O-antigens of gram-negative
bacteria



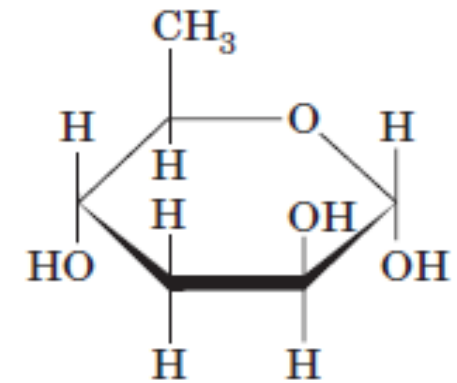
**2-Keto-3-deoxyoctanoate
(KDO)**



L-Glycero-D-mannoheptose



**Abequose
(Abe)**



Tyvelose

Gram positive bacteria

- Micrococcus
- Staphylococcus
- Streptococcus
- Leuconostoc

Gram negative bacteria

- E. coli
- Salmonella
- Klebsiella
- Shigella
- Pseudomonas

Differences between cell wall of Gram positive and Gram negative bacteria

Property of bacteria	Gram Positive	Gram Negative
Thickness of wall	20-80 nm	10 nm
Number of layers in wall	1	2
Peptidoglycan content	>50%	10-20%
Teichoic acid in wall	+	-
Lipid & lipoprotein content	0-3%	58%
Protein content	0%	9%
Lipopolysaccharide	0	13%
Sensitive to penicillin	Yes	Less sensitive
Digested by lysozyme	Yes	Weakly

PEPTIDOGLYCAN COMPOSITION

- An important component of the cell wall in bacteria is the peptidoglycan layer.
- The peptidoglycan layer, also known as murein, is a mesh-like polymer which is composed of many identical subunits.
- It consists of covalently linked polypeptide and polysaccharide chains. The polysaccharide component is made up of linear chains of alternating β (1 $\rightarrow$ 4) N-acetyl glucosamine and N-acetyl muramic acid. The polypeptide component is made of tetrapeptide side chains which consist of four alternating D and L amino acids. The tetrapeptides are connected to the lactic acid group of N-acetyl muramic acid through an amide bond.
- The tetrapeptide chains usually consist of L-Alanine, D-Glutamic acid, meso Diaminopimelic acid (can be substituted with another diamino acid like L-Lysine) and D-Alanine. The presence of D-Amino Acids protects the cell from the degradative action of peptidases which only recognize the L isomers of amino acids. The parallel tetrapeptide side chains of adjacent polysaccharide chains are joined by cross-links between the peptides. Often, the carboxyl group of the terminal D-Alanine is connected to the amino group of the diaminopimelic acid. However, a peptide interbridge may be present instead. Most gram-negative bacteria do not have the peptide interbridge. The peptidoglycan layer is a 20-80 nm thick homogenous layer in gram-positive bacteria whereas it is only 2-7 nm thick in gram-negative bacteria and more complex.

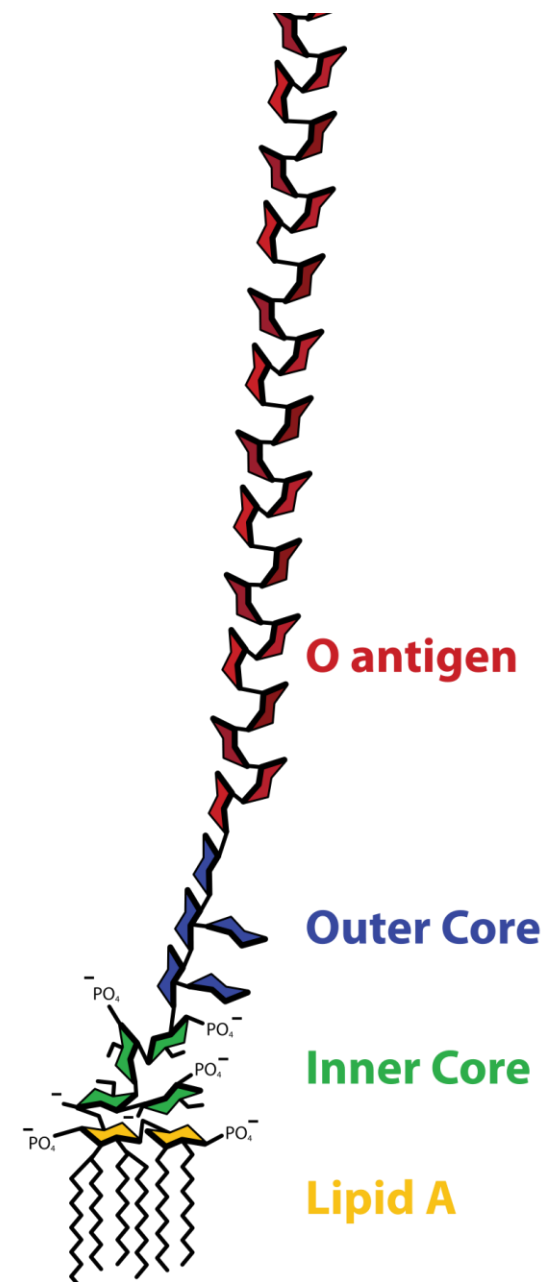
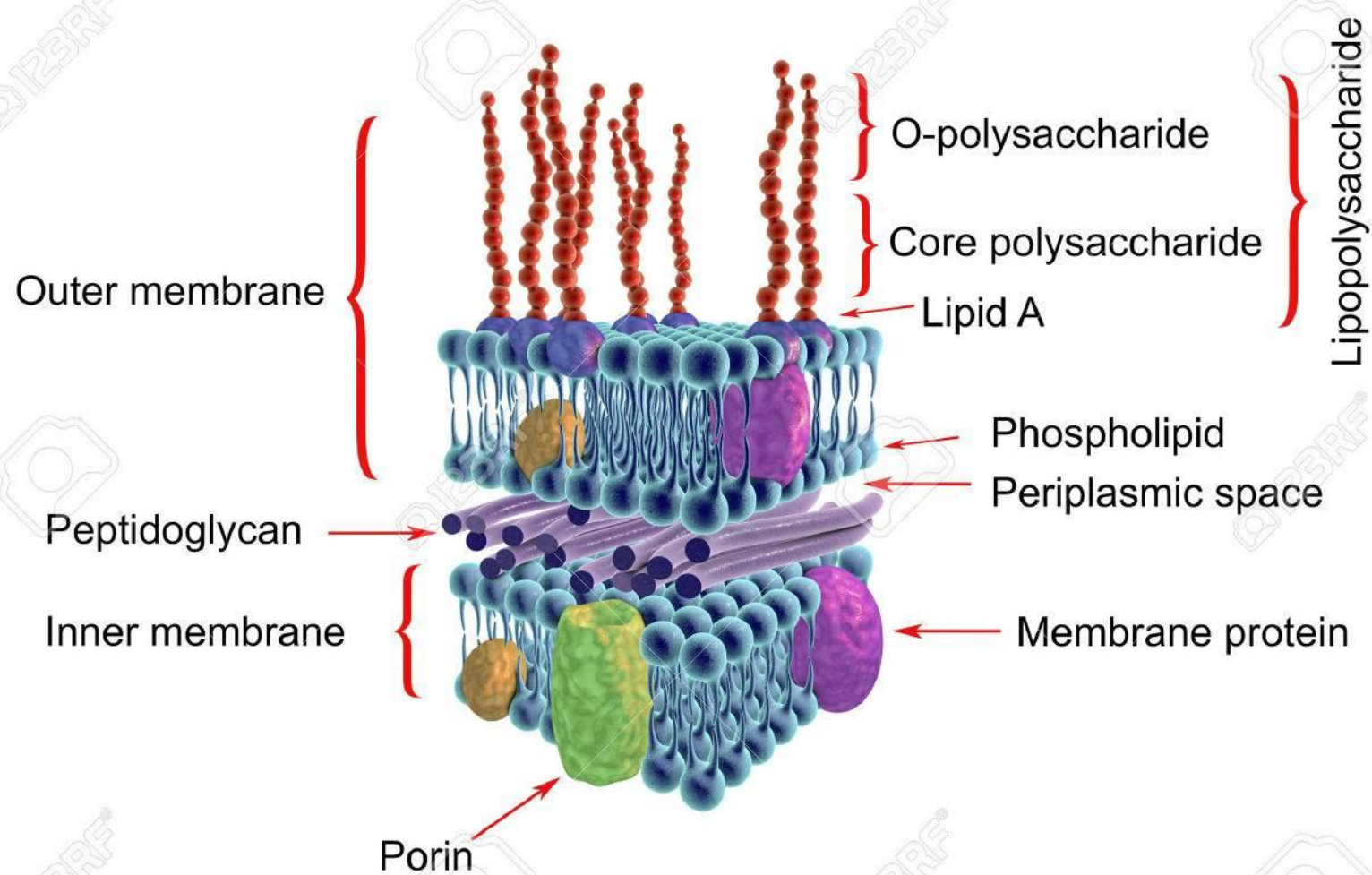
- **Lipopolysaccharides(LPS)-**
- Lipopolysaccharides (LPS) are also known as lipopolyglycans and endotoxins.
- They are large molecules consisting of a lipid and a polysaccharide composed of O-Antigen, outer core and inner core joined by a covalent bond.
- They are found in the outer membrane of the Gram Negative Bacteria.
- The term lipopolysaccharide "LOS" is used to refer to a low-molecular-weight form of bacterial lipopolysaccharides.

- **Structure-**

- Bacterial Lipopolysaccharides are macromolecules of molecular mass 10-20kDa made up of three structural components-
- O-Antigen- A repetitive glycan polymer contained within an LPS is referred to as the O-Antigen, O-polysaccharide or O side-chain. The O-Antigen is attached to the core oligosaccharide, and comprises the outermost domain of the LPS molecule. The presence or absence of O-Polysaccharide determines whether the LPS is considered rough or smooth. Full length O- chains would make the LPS smooth, whereas the absence or reduction of O-chains would make the LPS rough. It is present on the outer surface of the bacterial cell and as a result it acts as a target for recognition by host antibodies.
- Core Polysaccharide- The core always contain an oligosaccharide component that attaches directly to Lipid A and commonly contains sugars.
- Lipid A- These Hydrophobic fatty acid chains anchor the LPS into the bacterial membrane and the rest of the LPS projects from the cell surface. The lipid A domain is responsible for the toxicity of the Gram- Negative cell wall. When the bacterial cells are lysed by the immune system, fragments of membrane containing Lipid A are released into the circulation causing fever, diarrhea and possible fatal endotoxic shock (also called septic shock).

- **Functions-**

- LPS is the major component of the outer membrane of the Gram Negative Bacteria and contributes greatly to structural integrity.
- It protects the membrane from certain kinds of chemical attacks.
- LPS also increases the negative charge of the cell membrane and helps stabilize the overall membrane structure .



Teichoic Acid

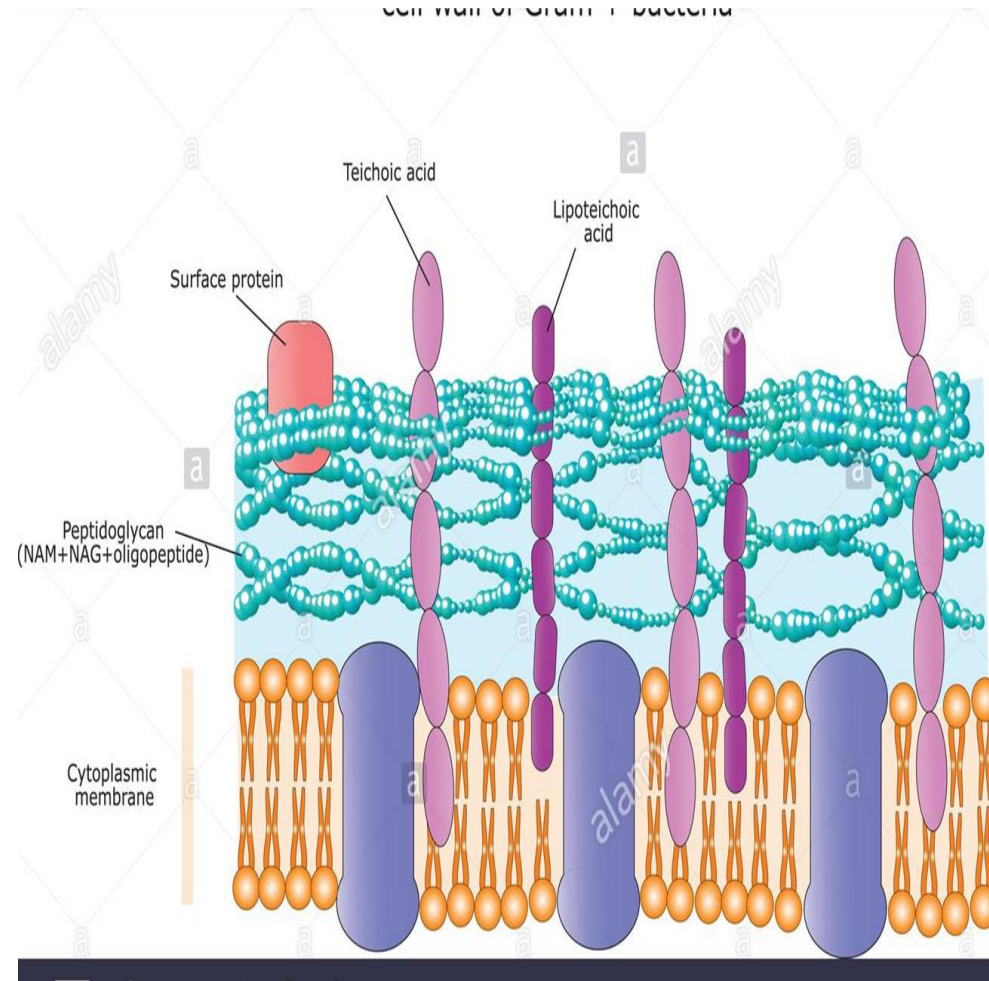
- Teichoic acid are bacterial copolymers of glycerol phosphate or ribitol phosphate and carbohydrate linked via phosphodiester bonds.
- Found within the cell wall of most Gram positive bacteria.
- E.g. *Staphylococcus*, *Streptococcus*, *Bacillus*, *Clostridium*, *Corynebacterium*, *Listeria* .
- Absent in Gram negative bacteria

- **Structure:**

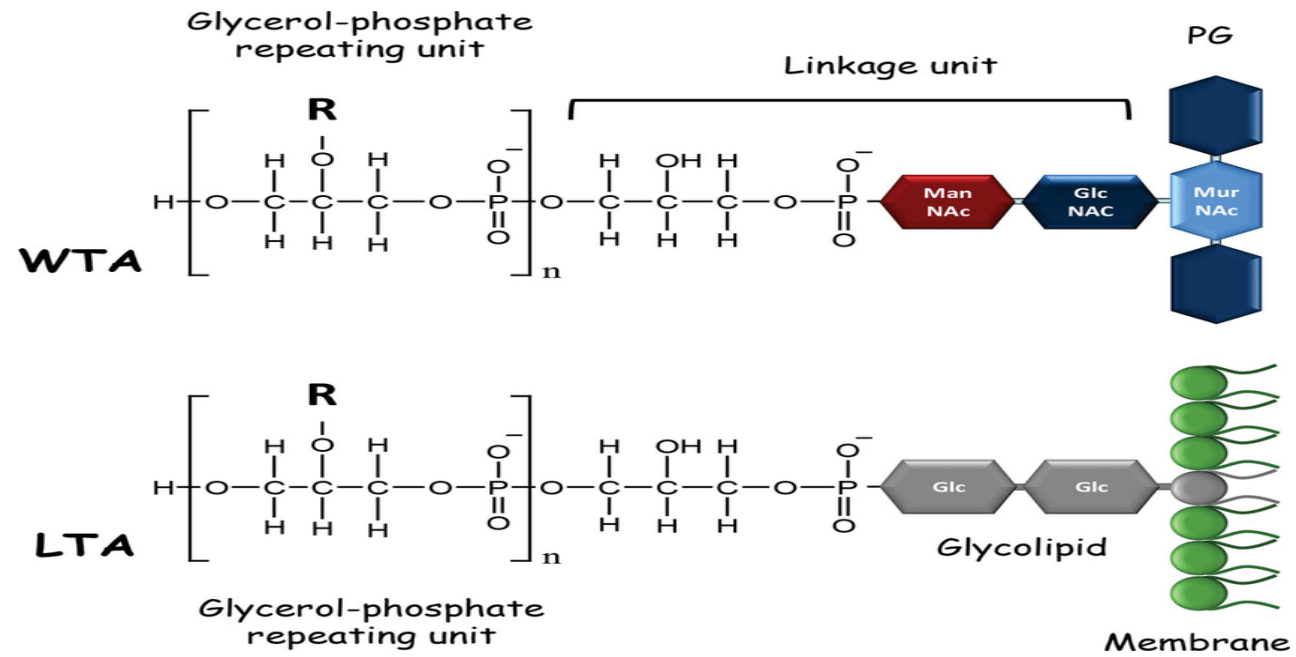
- teichoic acids that are anchored to the lipid membrane are referred to as lipoteichoic acids (LTAs), whereas teichoic acids that are covalently bound to peptidoglycan are referred to as wall teichoic acids (WTA).

- **Functions:**

- provides rigidity to the cell (by attracting cations such as Mg and Na)
- Cell shape determination
- Regulation in cell division (by limiting the ability of autolysins to break the beta(1-4) bond between the N-acetyl glucosamine and the N-acetyl glucosamine and N-acetylmuramic acid.
- Regulation of ion homeostasis
- Protection from host defence and antibiotics
- effects on adhesion and colonisation



Cell wall structure of Gram +ve bacteria



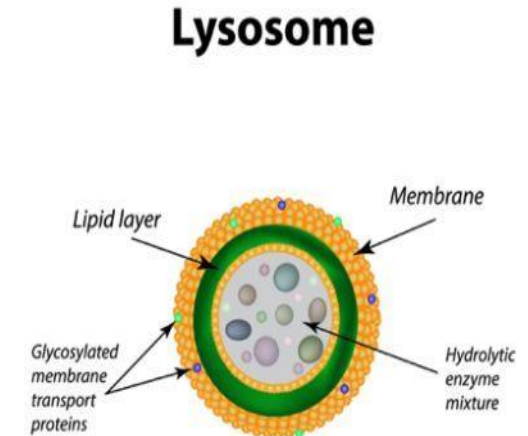
Schematic representation of wall teichoic acids and lipoteichoic acids

ENDOPLASMIC RETICULUM

- Endoplasmic reticulum is a continuous membrane system that forms a series of flattened sacs within the cytoplasm of eukaryotic cells and serves multiple functions, one of the most important is the synthesis, folding, modification and transport of proteins. All eukaryotic cells contain an endoplasmic reticulum. In animal cells, the endoplasmic reticulum usually constitutes more than half of the membranous content of the cell. Differences in certain physical and functional characteristics distinguish the two types of endoplasmic reticulum, known as rough endoplasmic reticulum and smooth endoplasmic reticulum.
- Rough endoplasmic reticulum is named for its rough appearance, which is due to the ribosomes attached to its outer cytoplasm surface. Rough endoplasmic reticulum lies immediately adjacent to the cell nucleus and its membrane is continuous with the outer membrane of the nuclear envelope. The ribosomes on the rough endoplasmic reticulum specialize in the synthesis of proteins that possess a signal sequence that directs them specifically to the endoplasmic reticulum. Proteins synthesized by the rough endoplasmic reticulum have specific destinations/locations. Some proteins, for example, remain within the endoplasmic reticulum whereas others are sent to the Golgi apparatus, which lies next to the endoplasmic reticulum. Proteins secreted from the Golgi apparatus are directed to lysosomes or to the cell membrane, still others are destined for secretion to the cell exterior. Proteins targeted for the Golgi apparatus are transformed from ribosomes on the rough endoplasmic reticulum into the lumen, which serves as the site of protein folding, modification and assembly.
- The proximity of the rough endoplasmic reticulum to the cell nucleus gives the endoplasmic reticulum unique control over protein processing. The rough endoplasmic reticulum is able to rapidly send signals to the nucleus when there is a problem/dysfunction in the protein synthesis and folding and thereby influences the overall rate of protein translation. When misfolded or unfolded proteins accumulate in the endoplasmic reticulum lumen, a signaling mechanism known as the unfolded protein response is activated. The response is adaptive, such that the UPR activation triggers reductions in protein synthesis and enhancements in the endoplasmic reticulum protein folding capacity and protein degradation. If the adaptive response fails, cells are directed to undergo apoptosis (programmed cell death).
- Smooth endoplasmic reticulum by contrast, is not associated with ribosomes and has different functions. It is involved in the synthesis of lipids, including cholesterol and phospholipids, which are used in the production of new cellular membrane. In certain cell types, smooth endoplasmic reticulum plays an important role in the synthesis of steroid hormones from cholesterol. In the cells of the liver, it contributes in the detoxification of drugs and harmful chemicals. The Sarcoplasmic reticulum is a specialized type of smooth endoplasmic reticulum that regulates the calcium ion concentration in the cytoplasm of striated muscles.

LYSOSOME

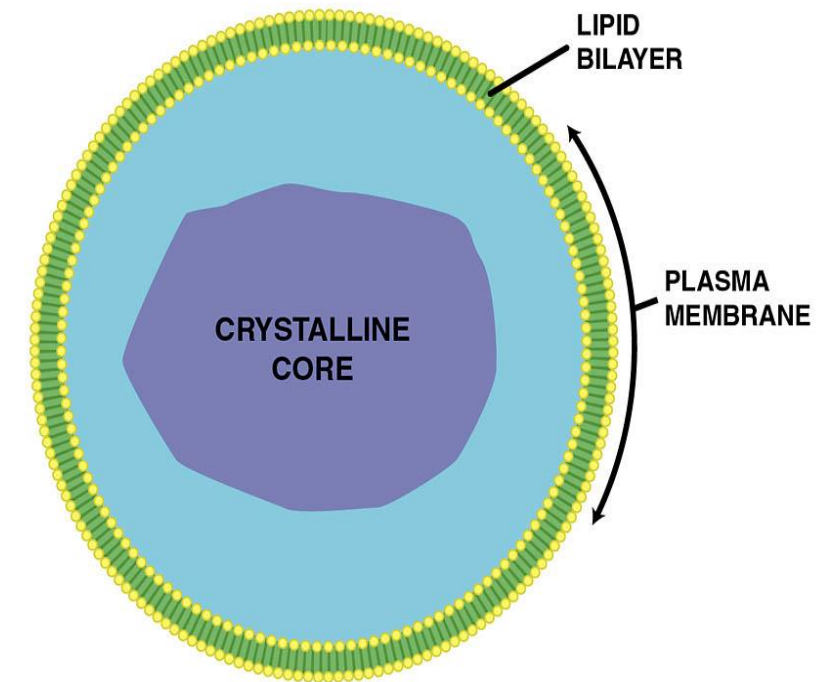
- Formation: Lysosomes originate by budding off from the membrane of the trans-Golgi network, a region of the Golgi Complex responsible for sorting newly synthesized proteins, which may be designated for use in lysosomes, endosomes, or the [plasma membrane](#).
- Structure: The lysosome is a specific type of organelle that's very acidic. So that means that it has to be protected from the rest of the inside of the cell. It's a compartment, then, that has a membrane around it that stores the digestive enzymes that require this acid, low-pH environment. Those enzymes are called hydrolytic enzymes, and they break down large molecules into small molecules. For example, large proteins into amino acids, or large carbohydrates into simple sugars, or large lipids into single fatty acids



- Function: The lysosomes fuse with membrane vesicles that derive from one of three pathways: endocytosis, auto phagocytosis, and [phagocytosis](#). In [endocytosis](#), extracellular macromolecules are taken up into the cell to form membrane-bound vesicles called endosomes that fuse with lysosomes. [Auto phagocytosis](#) is the process by which old organelles and malfunctioning cellular parts are removed from a cell; they are enveloped by internal membranes that then fuse with lysosomes. [Phagocytosis](#) is carried out by specialized cells (e.g., macrophages) that engulf large extracellular particles, such as dead cells or foreign invaders (e.g., [bacteria](#)), and target them for lysosomal [degradation](#). Many of the products of lysosomal digestion, such as [amino acids](#) and [nucleotides](#), are recycled back to the cell for use in the synthesis of new cellular components.

- PEROXISOME

- Formation: The assembly of peroxisomes is fundamentally similar to that of [mitochondria](#) and chloroplasts, rather than to that of the [endoplasmic reticulum](#), [Golgi apparatus](#), and lysosomes. Proteins destined for peroxisomes are translated on free cytosolic [ribosomes](#) and then transported into peroxisomes as completed [polypeptide](#) chains. Phospholipids are also imported to peroxisomes, via phospholipid transfer [proteins](#), from their major site of synthesis in the ER. The import of proteins and [phospholipids](#) results in [peroxisome](#) growth, and new peroxisomes are then formed by division of old ones.
- Structure: Peroxisomes are small (0.1-1 μm diameter) subcellular compartments (organelles) with a fine, granular matrix and surrounded by a single bio membrane which are located in the cytoplasm of a cell. Compartmentalization creates an optimized environment to promote various metabolic reactions within peroxisomes required to sustain cellular functions and viability of the organism.



- **Function:** Peroxisomes contain at least 50 different [enzymes](#), which are involved in a variety of biochemical pathways in different types of cells. Peroxisomes originally were defined as organelles that carry out oxidation reactions leading to the production of hydrogen peroxide. Because hydrogen peroxide is harmful to the cell, peroxisomes also contain the enzyme [catalase](#), which decomposes hydrogen peroxide either by converting it to water or by using it to oxidize another organic compound. A variety of substrates are broken down by such oxidative reactions in peroxisomes, including uric acid, amino acids, and [fatty acids](#). The oxidation of fatty acids is a particularly important example, since it provides a major source of metabolic energy. In animal cells, fatty acids are oxidized in both peroxisomes and [mitochondria](#), but in [yeasts](#) and plants fatty acid oxidation is restricted to peroxisomes.
- In addition to providing a compartment for oxidation reactions, peroxisomes are involved in lipid biosynthesis. In animal cells, [cholesterol](#) is synthesized in peroxisomes as well as in the [ER](#). In the liver, peroxisomes are also involved in the synthesis of bile acids, which are derived from cholesterol.
- peroxisomes in leaves are involved in **photorespiration**, which serves to metabolize a side product formed during [photosynthesis](#). CO_2 is converted to carbohydrates during photosynthesis via a series of reactions called the [Calvin cycle](#).